

Question Number	Answer	Additional Guidance	Mark
4a	A description that includes two of the following points: • find a suitable mass/age of leaves/concentration of extract/method of extraction (1) • find a suitable temperature/pH/medium/time/species of bacteria (1) • find suitable method to measure {inhibition/antibacterial effect} (1)	A method to provide quantitative results	Exp 2

Question Number	Answer	Additional Guidance	Mark
4b	An answer that includes nine of the following points: • clear statement of the dependent variable e.g. zone of inhibition (1) • description of method of preparation of extract (1) • method of preparing bacterial lawn/broth/pour plate (1) • method of applying extract (1) • detail of measuring dependent variable (1) • description of aseptic technique (1) • use of a control for comparison (1) • Incubate at stated temperature (1) • two variables that need to be controlled (1) • method of control of one named variable (1) • repeat the whole experiment to calculate (mean) and SD/error bars (1)	Ignore amount ALLOW different valid methods. Wells/discs/drop in broth/serial dilution/put on agar e.g. ruler/grid/different orientations ignore taping dishes But not more than 30° e.g. temperature/pH/incubation time/size of disc/medium/species of bacteria/source/mass of leaves accept to measure variability of data	9 exp

Question Number	Answer	Additional Guidance	Mark
4c	An answer that includes the following points: - table for collecting raw data with headings and units with means calculated from repeats (1) - bar graph format with labelled axes (1) - use of an appropriate statistical test (1)	accept with several concentrations description of mean calculated in text or mean on one graph label accept line graph if several concentrations used Accept a (named) correlation test with line graph a test for difference if bar graph	3 exp

Question Number	Answer	Additional Guidance	Mark
4d	An answer that includes two of the following points: - difficult to measure distances/diameters/ZOI with precision (1) - only tested against one species of bacteria (1) - contamination (1) - bacteria cultured in aerobic conditions and gut is anaerobic or not cultures at human body temperature (1)	Accept errors in use of colorimeter Accept the bacteria used might be resistant Accept conditions might not be aseptic Ignore ref to pathogens	2 exp

(Total for question 4 = 16 marks)

